GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Electromagnetics (EMFT) — Official GATE ECE Syllabus

Paper Code: EC • Section 8: Electromagnetics (8 - 10 Marks)

Weightage: 8 - 10 Marks • **Subject Code:** EMFT • **Key Focus:** Maxwell's Equations, Plane Waves, Transmission Lines, Waveguides, Antennas

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: ELECTROSTATICS & MAGNETOSTATICS

- Coordinate systems: Cartesian (x, y, z), Cylindrical (rho, phi, z), and Spherical (r, theta, phi). Differential length, surface, and volume vector elements. Gradient, divergence, and curl in all three coordinates.
- Divergence theorem (Gauss-Ostrogradsky theorem) and Stokes' theorem.
- Coulomb's Law, Electric field intensity (E), field due to continuous charge distributions (line, surface, and volume charge distributions).
- Electric flux density (D = epsilon * E), Gauss's Law (differential form: div D = rho_v, integral form: closed surface integral D.ds = Q_enclosed).
- Electrostatic potential (V), relationship between E and V (E = -grad V). Potential due to point charges and electric dipoles.
- Poisson's and Laplace's equations: div^2 V = -rho_v / epsilon (Poisson), div^2 V = 0 (Laplace). Uniqueness theorem.
- Electrostatic boundary conditions at material interfaces: Tangential E is continuous (E1t = E2t), Normal D is discontinuous by surface charge density (D1n - D2n = rho_s; if charge-free interface D1n = D2n).
- Capacitance calculations: Parallel plate, coaxial cylindrical, and concentric spherical capacitors. Energy stored in electrostatic field: W_E = 0.5 * integral(D.E dv) = 0.5 * C * V^2.
- Biot-Savart Law for line, surface, and volume currents.
- Magnetic field intensity (H) and magnetic flux density (B = mu * H). Ampere's Circuital Law (differential form: curl H = J, integral form: closed line integral H.dl = I_enclosed).
- Magnetic vector potential (A where B = curl A), Coulomb gauge condition (div A = 0).
- Magnetostatic boundary conditions: Tangential H is discontinuous by surface current density (H1t - H2t = K x a_n; if current-free H1t = H2t), Normal B is always continuous (B1n = B2n, non-existence of isolated magnetic monopoles).
- Inductance and mutual inductance calculations. Energy stored in magnetic field: W_M = 0.5 * integral(B.H dv) = 0.5 * L * I^2.

SECTION: MAXWELL'S EQUATIONS & TIME-VARYING FIELDS

- Faraday's Law of Electromagnetic Induction: Transformer EMF and Motional EMF. Differential form: curl E = -dB/dt, Integral form: closed line integral E.dl = -d(phi)/dt.
- Inconsistency of Ampere's Law for time-varying fields: Equation of continuity (div J = -d(rho_v)/dt).
- Maxwell's displacement current density: Jd = dD/dt.
- Complete Maxwell's Equations in differential and integral forms:
- 1. div D = rho_v (Gauss's Law for Electric fields)
- 2. div B = 0 (Gauss's Law for Magnetic fields)
- 3. curl E = -dB/dt (Faraday's Law)
- 4. curl H = J + dD/dt (Ampere-Maxwell Law)
- Time-harmonic (sinusoidal steady-state) Maxwell's equations using phasor notation.
- Poynting Theorem: Energy conservation in electromagnetic fields. Instantaneous Poynting vector: S = E x H (W/m^2).

• Time-average Poynting vector for time-harmonic fields: S_avg = 0.5 * Re{E x H*} (W/m^2).

SECTION: UNIFORM PLANE WAVES & WAVE PROPAGATION

• Electromagnetic wave equations for E and H derived from Maxwell's curl equations (Helmholtz wave equations in phasor form).

• Uniform plane wave propagation in lossless dielectric media: Propagation constant (gamma = alpha + j*beta = j*omega*sqrt(mu*epsilon)). Phase constant beta = omega*sqrt(mu*epsilon) = 2*pi / lambda.

• Phase velocity (vp = omega / beta = 1 / sqrt(mu*epsilon) = c / sqrt(epsilon_r*mu_r)), Intrinsic impedance (eta = sqrt(mu / epsilon); for free space eta0 = 120*pi ~ 377 Ohms).

• Wave propagation in lossy, conducting media: Complex permittivity (epsilon_c = epsilon - j*sigma/omega), loss tangent (tan delta = sigma / (omega * epsilon)).

• Good dielectrics (loss tangent << 1) vs. Good conductors (loss tangent >> 1):

• - Good conductors: Attenuation constant alpha = beta = sqrt(pi * f * mu * sigma).

• - Skin depth (depth of penetration): delta = 1 / alpha = 1 / sqrt(pi * f * mu * sigma) (distance over which wave amplitude decays to 1/e ~ 36.8% of surface value).

• Wave polarization: Direction of electric field vector orientation in space as a function of time. Linear, Circular (Right Hand Circular and Left Hand Circular), and Elliptical polarization. Polarization criteria based on phase difference and component amplitudes.

• Reflection and refraction of plane waves: Normal incidence at plane boundary between two media. Reflection coefficient (Gamma = (eta2 - eta1) / (eta2 + eta1)), Transmission coefficient (tau = 2*eta2 / (eta2 + eta1) = 1 + Gamma).

• Standing wave ratio: SWR = (1 + |Gamma|) / (1 - |Gamma|).

• Oblique incidence of plane waves: Snell's Law of Reflection and Snell's Law of Refraction (n1*sin theta1 = n2*sin theta2). Parallel (TM) and perpendicular (TE) polarization.

• Brewster Angle (polarizing angle, total transmission, Gamma = 0): tan theta_B = sqrt(epsilon2 / epsilon1) for parallel polarization. Critical Angle (total internal reflection occurs when wave travels from denser to rarer medium theta_c = arcsin(sqrt(epsilon2 / epsilon1))).

SECTION: TRANSMISSION LINES & SMITH CHART

• Distributed parameter transmission line model: Resistance (R in Ohms/m), Inductance (L in H/m), Conductance (G in S/m), Capacitance (C in F/m).

• Telegrapher's equations in time and frequency domains.

• Propagation constant: gamma = sqrt((R + j*omega*L)*(G + j*omega*C)) = alpha + j*beta.

• Characteristic impedance: Z0 = sqrt((R + j*omega*L) / (G + j*omega*C)).

• Lossless transmission lines (R = 0, G = 0): Characteristic impedance Z0 = sqrt(L / C) (purely resistive), phase constant beta = omega * sqrt(L * C), phase velocity vp = 1 / sqrt(L * C).

• Distortionless transmission line condition (Heaviside condition): R / L = G / C. Results in frequency-independent attenuation (alpha = sqrt(R * G)) and linear phase (beta = omega*sqrt(L*C)), eliminating dispersion.

• Finite transmission lines terminated with load Z_L: Input impedance formula: Zin(l) = Z0 * [(ZL + j*Z0*tan(beta*l)) / (Z0 + j*ZL*tan(beta*l))].

• Special transmission line lengths:

• - Quarter-wave transformer (l = lambda / 4): Zin = Z0^2 / ZL (used for impedance matching between two resistive lines).

• - Half-wave line (l = lambda / 2): Zin = ZL (repeats load impedance every half wavelength).

• - Short-circuited line (ZL = 0): Zin = j*Z0*tan(beta*l) (acts as pure reactance, inductors or capacitors depending on length).

• - Open-circuited line (ZL = infinity): Zin = -j*Z0*cot(beta*l).

• Voltage reflection coefficient at load: Gamma_L = (ZL - Z0) / (ZL + Z0).

• Voltage Standing Wave Ratio: VSWR (s) = V_max / V_min = (1 + |Gamma|) / (1 - |Gamma|). Range: 1 <= s <= infinity.

• Smith Chart: Graphical tool mapped from complex impedance plane to reflection coefficient Gamma plane. Constant resistance circles, constant reactance circles. Stub matching: Single stub and double stub tuning.

SECTION: WAVEGUIDES & RESONANT CAVITIES

• Rectangular waveguides: Physical structure, electromagnetic boundary conditions (tangential E = 0 on conducting walls).

• Transverse Electric (TE_mn) and Transverse Magnetic (TM_mn) waveguide modes. Impossibility of TEM mode in hollow single-conductor waveguides.

• Cutoff frequency formula for TE_mn and TM_mn modes: fc_mn = (c / 2) * sqrt((m / a)^2 + (n / b)^2) where a is width and b is height (a > b).

• Dominant mode: Mode with the lowest cutoff frequency. For rectangular waveguide with a > b, dominant mode is TE10 (fc_10 = c / (2*a)). Degenerate modes.

• Guide wavelength: lambda_g = lambda_0 / sqrt(1 - (fc / f)^2) > lambda_0.

• Phase velocity (vp = c / sqrt(1 - (fc / f)^2) > c) vs. Group velocity (vg = c * sqrt(1 - (fc / f)^2) < c). Relationship: vp * vg = c^2.

• Wave impedance: Z_TE = eta0 / sqrt(1 - (fc / f)^2) and Z_TM = eta0 * sqrt(1 - (fc / f)^2). Attenuation in waveguides due to wall conductor losses and dielectric losses.

• Rectangular resonant cavities: Resonant frequency formula fres_mnp = (c / 2) * sqrt((m/a)^2 + (n/b)^2 + (p/d)^2). Dominant cavity mode.

SECTION: ANTENNAS & RADIATION

• Radiation mechanism: Accelerated charges, retarded vector potential A(r, t).

• Infinitesimal dipole (Hertzian dipole, length dl << lambda): Radiation fields (E_theta and H_phi in far-field proportional to 1/r), Radiation resistance R_rad = 80*pi^2*(dl / lambda)^2.

• Half-wave dipole antenna (length l = lambda / 2): Current distribution (sinusoidal, zero at ends, maximum at center), radiation resistance R_rad = 73 Ohms, input impedance ~ 73 + j42.5 Ohms.

• Quarter-wave monopole antenna (over infinite ground plane, image theory): Radiation resistance R_rad = 36.5 Ohms.

• Antenna parameters: Radiation pattern, Radiation power density, Radiation intensity U(theta, phi) = r^2 * S_avg.

• Directivity: D = 4*pi * U_max / P_rad = 4*pi / Omega_A. For Hertzian dipole D = 1.5 (1.76 dBi); for half-wave dipole D = 1.64 (2.15 dBi).

• Antenna efficiency (eta_rad = R_rad / (R_rad + R_loss)), Antenna gain (G = eta_rad * D).

• Effective aperture (Ae): Relationship with directivity Ae = (lambda^2 / (4*pi)) * D.

• Friis Transmission Formula: Pr / Pt = Gt * Gr * (lambda / (4*pi*R))^2.

• Radar Range Equation: Pr = (Pt * Gt * Gr * lambda^2 * sigma) / ((4*pi)^3 * R^4).

• Antenna arrays: Principle of pattern multiplication (Total Field Pattern = Element Pattern * Array Factor). Two-element array, N-element uniform linear array (broadside vs. end-fire arrays).

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

• Elements of Electromagnetics — Matthew N.O. Sadiku (Oxford University Press)
• Electromagnetic Waves and Radiating Systems — Edward C. Jordan & Keith G. Balmain
• Antenna Theory: Analysis and Design — Constantine A. Balanis (Wiley)

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

• TEM waves cannot propagate in a hollow single-conductor waveguide because closed line integral of H would violate Ampere's law with zero enclosed current.
• In wave propagation through conductors, skin depth delta = sqrt(1 / (pi*f*mu*sigma)). High frequency RF currents travel strictly on the outer perimeter.
• Input impedance of quarter-wave transformer: Zin = Z0^2 / ZL. If ZL is open circuit, Zin is short circuit (0 Ohms).